



AIMS05-01-004

Issue 3

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AIMS
Airbus Material Specification

Carbon fibre reinforced prepreg

5HS fabric / 280g/m² / 180°C curing class

Standard modulus fibre

Material Specification

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1 Scope

This Material Specification defines the requirements of carbon fiber reinforced five harness satin fabric prepreg, 280g/m², curing at 180°C, for aerospace application.

Unless otherwise specified by the drawing, the order or the inspection schedule, the present specification shall be used in conjunction with the referenced documents.

2 Application

Manufacture of structural components, temperature range -55°C to +95°C. Materials to be delivered according to the relevant ABS, cured ply thickness 0.28 mm ± 0.015 mm, fibre volume fraction (56 ± 5) %.

Restrictions concerning the application, if applicable

The requirements specified in this Material Specification shall not be used as stress allowables.

3 Normative references

This Airbus specification incorporates by dated or undated reference provisions from other publications. All normative references cited at the appropriate places in the text are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this Airbus specification only when incorporated in it by amendment of revision. For undated references, the latest issue of the publication referred to shall be applied.

ABS 5003	Airbus Standard - EP-Prepreg Woven carbon fibre fabrics preimpregnated with epoxy resin
AIMS 05-01-000 Part 1	Airbus Material Specification - Technical Specification - Carbon fibre reinforced epoxy prepreg - General requirements
AIMS 05-01-000 Part 3	Airbus Material Specification - Technical Specification - Carbon fibre reinforced epoxy prepreg - Qualification and batch release testing Fabric / 180°C-Curing class
AIMS05-01-000 Part 100	Carbon fiber reinforced epoxy prepreg Structural material Acceptance Testing and Storage Life Prolongation
AITM 1.0001	Airbus Test Method - Fibre reinforced plastics. Determination of mechanical degradation due to chemical paint strippers
AITM 1.0002	Airbus Test Method - Fibre reinforced plastics. Determination of in-plane shear properties
AITM 1.0003	Airbus Test Method - Fibre reinforced plastic materials. Determination of the glass transition temperatures
AITM 1.0005	Airbus Test Method - Carbon fibre reinforced plastics. Determination of interlaminar fracture toughness energy. Mode I
AITM 1.0006	Airbus Test Method - Carbon fibre reinforced plastics. Determination of interlaminar fracture toughness energy. Mode II
AITM 1.0007	Airbus Test Method - Fiber reinforced plastics. Determination of plain, open hole and filled hole tensile strength
AITM 1.0008	Airbus Test Method - Fiber reinforced plastics. Determination of plain, open hole and filled hole compression strength
AITM 1.0009	Airbus Test Method - Fibre reinforced plastics. Determination of bearing strength
AITM 1.0010	Airbus Test Method - Fibre reinforced plastics. Determination of compression strength after impact
AITM 3.0008	Airbus Test Method - Determination of the extent of cure by differential scanning calorimetry
ASTM E831	Standard test method for linear thermal expansion of solid materials by thermomechanical analysis

EN 2378	Aerospace series. Fibre reinforced plastics. Determination of water absorption by immersion
EN 2557	Aerospace series. Carbon fibre preimpregnates. Determination of mass per unit area
EN 2558	Aerospace series. Carbon fibre preimpregnates. Determination of the percentage of volatile matter.
EN 2559	Aerospace series. Carbon fibre preimpregnates. Determination of the percentage of resin and fibre content and the mass of fibre per unit area.
EN 2560	Aerospace series. Carbon fibre preimpregnates. Determination of the resin flow.
EN 2561	Aerospace series. Carbon fibre reinforced plastics unidirectional laminates. Tensile test parallel to the fibre direction
EN 2563	Aerospace series. Unidirectional laminates. Determination of the apparent interlaminar shear strength
EN 2597	Aerospace series. Unidirectional laminates carbon thermosetting resin. Tensile test perpendicular to the fibre direction
EN 2823	Aerospace series. Fibre reinforced plastics. Determination of the effect of exposure to humid atmosphere on physical and mechanical characteristics
EN 2850	Aerospace series. Carbon fibre thermosetting resin unidirectional laminates compression test parallel to the fibre direction
EN 3615	Aerospace series. Fibre Reinforced plastics. Determination of the conditions of exposure to humid atmosphere and moisture absorption
ISO 10119	Carbon fibres. Determination of density
ISO 1183	Plastics. Methods for determining the density and relative density of non-cellular plastics

4 Technical requirements

4.1 General

Specification values are minimum except where stated.

4.1.1 Material description

See table 1.

4.1.2 Worklife and storage conditions

See table 2.

4.1.3 Processing

See table 3.

4.2 Specific

4.2.1 Physical-chemical properties

Uncured and cured properties, see table 4.

4.2.2 Mechanical properties

See table 5.

5 Designation

See relevant ABS5003.

6 Technical specification

The general technical requirements and responsibilities corresponding to this Material Specification are described in the Technical Specification AIMS05-01-000 Part 1 and Part 3.

7 Incoming inspection

Materials qualified against this AIMS shall undergo "Technical Incoming Inspection" as per AIMS05-01-000 Part 100 prior to use.

Table 1: Material description

Characteristic	Requirements
Material type	5HS carbon fiber fabric/epoxy resin
Formulation	Fiber: standard modulus carbon fiber Resin: Epoxy, toughened
Form and method of production	Prepreg, cured in an autoclave at 180°C
Technical Specification	AIMS 05-01-000 Part 1 and Part 3
Resin content by weight, uncured	37% nominal
Service temperature range	-55°C up to 95°C

Table 2: Worklife and storage conditions

Properties	Symbol	Unit	Requirements
Worklife at 30°C/75% R.H.	-	days	> 10
Storage life at -18°C	-	months	> 12
Worklife at (25+2)°C/(45+10)% RH	-	days	> 15

Table 3: Processing conditions

Cure type	Autoclave		
Cure conditions	Heating rate	°C/min	0.2 – 3.5
	Cure temperature	°C	175 – 190
	Cure time	min	max. 210
	Cure pressure	MPa	0.35 – 0.72
	Cooling rate	°C/min	0.2 – 3.5
	Vacuum (abs)	MPa	0.005 – 0.1

Table 4: Physical properties

Properties	Test methods	Unit	Requirements
Uncured			
Prepreg areal weight	EN 2557	g/m ²	444 + 25
Fiber areal weight	EN 2559	g/m ²	280 + 12
Resin density	ISO 1183 Method A	g/cm ³	1.30 + 0.05
Fiber density	ISO 10119	g/cm ³	1.78 + 0.04
Volatile content	EN 2558	%	< 2
Resin content	EN 2559	%	37 + 3
Physic-chemical tests	AIMS 05-01-000 part 1	-	See relevant IPS
Resin flow	EN 2560	%	See relevant IPS
Tack	To be agreed between manufacturer and Airbus/Natco	-	-
Cured			
Coefficient of thermal expansion	ASTM E831	(°C)-1	To be reported
Water uptake	EN 3615, 70°C/85% R.H.	%	< 1.5
Water uptake	EN 2378, 70°C/water	%	< 1.9
Extent of cure	AITM 3-0008	%	> 90
Note: - The stated requirements are average values.			

Table 5: Mechanical properties

Properties	Test method	Unit	Conditioning (2)	Test temp. [°C]	Requirements (1)	
					X	Xmin
Material Code					AIMS05-01-004	
Glass transition temperature, Tgonset	AITM 1.0003	°C	dry	-	-	150
			wet.	-	-	120
Tensile strength (warp)	EN 2561 type B	MPa	dry	-55	700	650
				RT	700	650
				70	700 ⁷⁾	650 ⁷⁾
				120#	700	650
			wet	70	700	650
				90	700	650
				120#	700	650
Tensile modulus (warp)	EN 2561 type B	GPa	dry	-55	65	60
				RT	65	60
				70	65 ⁷⁾	60 ⁷⁾
				120#	65	60
			wet	70	65	60
				90	65	60
				120#	65	60
Tensile strength (weft)	EN 2597 type B	MPa	dry	-55	700	650
				RT	700	650
				120#	700	650
			wet	70	700	650
				90	700	650
				120#	700	650
(continued)						

(continued)

Table 5: Mechanical properties

Properties	Test method	Unit	Conditioning ²⁾	Test temp. [°C]	Requirements ¹⁾ X Xmin	
Material Code					AIMS05-01-004	
Tensile modulus (weft)	EN 2597 type B	GPa	dry	-55	65	60
				RT	65	60
				120#	65	60
			wet	70	65	60
				90	65	60
				120#	65	60
Compression strength (warp)	EN 2850 type B	MPa	dry	-55	860	800
				RT	860	800
				70	800 ⁷⁾	730 ⁷⁾
				120#	700	650
			wet	70	700	625
				90	600	575
				120#	545	450
				Compression modulus (warp)	EN 2850 type B	GPa
RT	57	52				
70	57	52				
120#	57	52				
wet	70	57	52			
	90	57	52			
	120#	57	52			
	RT	780	670			
wet	120#	650	540			
	70	650	540			
	90	515	460			
	120#	475	425			
wet	RT	57	52			
	120#	57	52			
	70	57	52			
	90	57	52			
wet	120#	57	52			
	70	57	52			
	90	57	52			
	120#	57	52			
In plane shear strength	AITM 1.0002	MPa	dry	-55	100	90
				RT	100	90
				70	95 ⁷⁾	85 ⁷⁾
				120	80	70
			wet	RT	100	90
				70	88	80
				90	70	65
				120	60	55
In plane shear modulus	AITM 1.0002	GPa	dry	-55	5.2	4.7
				RT	4.4	4.3
				70	3.9 ⁷⁾	3.6 ⁷⁾
				120	3.3	2.9
			wet	RT	4.2 ⁷⁾	4.1 ⁷⁾
				70	3.8	3.5
				90	3.2	2.8
				120	2.1	1.6
(continued)						

(continued)

Table 5: Mechanical properties

Properties	Test method	Unit	Conditioning 2)	Test temp. [°C]	Requirements 1) X Xmin	
Material Code					AIMS 05-01-004	
Interlaminar shear strength	EN 2563	MPa	dry	-55	65	58
				RT	70	60
				70	65 7)	55 7)
				120	53	48
			wet	RT	60 7)	50 7)
				70	51	45
				90	45	40
				120	32	30
			water	70	49	45
Compression after impact 3)	AITM 1.0010	MPa	dry	RT	220	200
			wet	70	4)	4)
G1C	AITM 1.0005	J/m ²	dry	RT	4)	4)
			wet	RT	4)	4)
G2C	AITM 1.0006	J/m ²	dry	RT	4)	4)
			wet	RT	4)	4)
Resistance to aggressive fluids						
Interlaminar shear strength (reference)	EN 2563	MPa	dry	RT	70	60
			wet	90	45	40
Interlaminar shear strength	EN 2563	MPa	(skydrol, 70°C/1000 h)	90	55 7)	50 7)
Interlaminar shear strength	EN 2563	MPa	(skydrol/water, 50°C/1000 h)	90	40 7)	35 7)
Interlaminar shear strength	EN 2563	MPa	(fuel,RT/1000h)	90	50 7)	45 7)
Interlaminar shear strength	EN 2563	MPa	(MEK, RT/1 h)	RT	70	60
In plane shear strength (paint strippers.Stripers to be defined in the qualification test programme)	AITM 1.0001	MPa	dry	RT	(4)	(4)
			wet	70#	(4)	(4)
Tension strength (unnotched) 5)	AITM 1.0007 A quasi-isotropic lay-up	MPa	dry	-55	510	460
				RT	510	460
				70	510	460
			wet	70	510	460
Open hole tension strength (notched) 5) 6)	AITM 1.0007 A quasi-isotropic lay-up	MPa	dry	-55	275	250
				RT	275	250
			wet	70	275	250
Filled hole tension strength (notched) 5) 6)	AITM 1.0007 A quasi-isotropic lay-up	MPa	dry	RT	300 7)	275 7)
				70	275 7)	250 7)
			wet	70	275 7)	250 7)
Compression strength (unnotched) 5)	AITM 1.0008 A quasi-isotropic lay-up	MPa	dry	RT	460 7)	415 7)
				70	445	400
			wet	70	460	415
Open hole compression strength (notched) 5) 6)	AITM 1.0008 A quasi-isotropic lay-up	-	dry	-55	290 7)	275 7)
				RT	290 7)	275 7)
			wet	70	225 7)	200 7)
Filled hole compression strength (notched) 5) 6)	AITM 1.0008 A quasi-isotropic lay-up	MPa	dry	RT	375	335
				70	370	330
			wet	70	370	330
(continued)						

(continued)

Table 5: Mechanical properties (concluded)

Properties		Test method	Unit	Conditioning (2)	Test temp. [°C]	Requirements (1)	
						X	Xmin
Material Code						AIMS05-01-004	
Bearing strength	2% elong	AITM 1.0009 A quasi-isotropic lay-up	MPa	dry	-55	620 ⁷⁾	560 ⁷⁾
					RT	590	530
				wet	70	480	430
					120	415	370
	Ult.			dry	-55	685 ⁷⁾	615 ⁷⁾
					RT	650	585
				wet	70	500	450
					120	425	380

Note: # special requirement for Airbus Spain, additional to requirement of AIMS 05-01-000 Part 3.

- ¹⁾ The stated requirements are average
- X means minimum average value.
 - Xmin means minimum individual value.
- All tension and compression test results are normalised to nominal thickness. Notched results calculated on gross area basis. Shear, bearing, ILSS, CAI requirements based on actual thickness.
- ²⁾ Conditioning "wet" is 70°C/85% Relative Humidity according to EN2823 and "water" is 70°C/water according to EN2378.
- ³⁾ Impact energy 20 J, approximately 0.3 mm indentation
- ⁴⁾ Values to be reported during the qualification
- ⁵⁾ Specimens width: 38 mm.
- ⁶⁾ Requirements using the total width of the specimens, without discounting the hole diameter
- ⁷⁾ Values to be checked/confirmed during qualification

RECORD OF REVISIONS

Issue	Clause modified	Description of modification
1 01/00		New standard
2 11/05	Tables 4 and 5	Update of material requirements
3 07/13	Chapter 3 Chapter 7	AIMS05-01-000 part 100 added Chapter 7 defined in line with Incoming Inspection (Part 100) implementation